

ADVANCED TECHNOLOGICAL INSTITUTE
HND in Information Technology



GH+

GOVERNMENT HOSPITAL PLUS

*A Digital Medical Records Management System for Government Hospitals of
Sri Lanka*

A PROPOSAL PRESENTED BY

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CHAPTER 1 – INTRODUCTION

1.1 PROBLEM STATEMENT

The healthcare network of government hospitals. This manual system creates significant challenges: patient records are frequently lost or damaged, duplicate records are created when patients visit different hospitals, critical medical histories are unavailable in emergencies, and there is no secure and centralized way for authorized medical professionals to access patient data. Furthermore, patients themselves have no reliable access to their own medical history, prescriptions, or diagnostic documents.

The absence of a digital, role-based medical records management system results in delayed diagnoses, medication errors, and overall inefficiency in patient care delivery. There is a clear and urgent need for a secure, centralized, and accessible digital health records platform specifically designed for Sri Lanka's government hospital network.

1.2 FUNCTIONAL REQUIREMENTS AND OBJECTIVES

The primary objective of the Government Hospital Plus (GH+) system is to develop a fully functional, web-based medical records management platform that digitizes patient health records for Sri Lanka's government hospitals. The system must meet the following functional requirements:

- Provide a secure, NIC-based login system for three distinct user roles: Patient, Doctor/Medical Staff, and Administrator.
- Allow patients to register online and submit their details for administrative approval before gaining access.
- Enable medical staff to search patients by NIC, name, city, or blood type and view complete medical histories.
- Allow doctors to upload new medical records including diagnosis, prescriptions, notes, and attached files (images, PDFs, audio, video).
- Enable patients to view their own medical history, prescriptions, and documents in a read-only mode.

- Allow patients to download individual medical records as a formatted PDF document.
- Provide the administrator with full system control including user approval, editing, deletion, audit logging, and reporting.
- Ensure mobile responsiveness so the system is fully accessible on smartphones and tablets.
- Automatically calculate and display patient age from date of birth throughout the system.
- Maintain a complete audit log of all user actions for security and accountability purposes.

CHAPTER 2 – LITERATURE REVIEW AND STUDY OF SIMILAR APPLICATIONS

2.1 LITERATURE REVIEW

2.1.1 Electronic Health Record (EHR) Systems

Electronic Health Record systems have been extensively studied as a means of improving healthcare delivery globally. Research consistently demonstrates that EHR adoption reduces medical errors, improves continuity of care, and enhances patient safety. Studies by the World Health Organization (WHO) highlight that digitizing health records in developing countries significantly reduces duplicate testing and improves diagnostic accuracy, particularly in emergency situations where historical patient data is critical.

2.1.2 Role-Based Access Control in Healthcare

Role-Based Access Control (RBAC) is a widely adopted security model in healthcare information systems. Literature emphasizes that strict role separation — ensuring patients access only their own data, doctors access patient records within their care, and administrators maintain system oversight — is essential to compliance with medical privacy standards. The principle of least privilege ensures that each user can only perform actions appropriate to their role, minimizing the risk of data breaches.

2.1.3 Web-Based Medical Systems in Developing Countries

Studies on web-based healthcare systems in developing countries indicate that PHP/MySQL stacks running on local servers (such as XAMPP) are particularly well suited for environments where internet connectivity may be unreliable or where cloud infrastructure is cost-prohibitive. Server-side rendering with PHP provides broad compatibility across devices and does not require high-end client hardware, making it ideal for deployment in government hospital settings in Sri Lanka.

2.1.4 Patient Data Privacy and Security

Medical data privacy is a critical concern in any healthcare system. Research in health informatics stresses the importance of encrypted password storage,

session-based authentication, input sanitization to prevent SQL injection attacks, and comprehensive audit logging. These mechanisms together ensure that patient data remains confidential and that all access and modifications to the system are traceable and accountable.

2.2 STUDY OF SIMILAR APPLICATIONS

2.2.1 HMIS – Hospital Management Information System (Sri Lanka)

The existing HMIS used in some Sri Lankan government hospitals is primarily an administrative tool focused on bed management and outpatient statistics. It lacks a patient-facing portal, does not support multi-format file uploads, and provides no mechanism for patients to directly access their own records. GH+ addresses all of these gaps by providing a dedicated patient dashboard with full read access to personal health histories.

2.2.2 MyCareHub (Australia)

MyCareHub is an Australian government digital health initiative that allows patients to access their medical records online. While comprehensive, it relies on expensive cloud infrastructure and a high level of internet penetration that is not universally available in Sri Lanka. GH+ is designed to operate on a local server environment, making it practical and cost-effective for implementation in Sri Lanka's government hospital context.

2.2.3 OpenMRS (Open Medical Record System)

OpenMRS is an open-source medical record system widely deployed across developing countries. It is highly customizable but requires significant technical expertise to deploy and maintain. GH+ offers a simpler, purpose-built alternative tailored specifically to Sri Lanka's government hospital structure, with an intuitive interface that requires minimal training for medical staff and patients alike.

2.3 MOTIVATION

The motivation for developing GH+ stems from direct observation of the inefficiencies within Sri Lanka's government healthcare system. Paper records are routinely lost when patients transfer between hospitals, prescriptions are hand-written and prone to misinterpretation, and there is no mechanism for

patients to independently verify or retrieve their medical history. By developing a locally hosted, role-based digital records system, GH+ aims to meaningfully improve patient care outcomes, reduce administrative burden on medical staff, and empower patients with transparent access to their own health data.

CHAPTER 3 – MATERIALS AND METHODS

3.1 MATERIALS

3.1.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	50 GB free disk space for database and file uploads
Network	Local Area Network (LAN) for multi-user access
Display	1366 × 768 resolution or higher
Input Devices	Keyboard, mouse, and optional scanner for documents

3.1.2 Software Requirements

Software	Purpose	Version / Note
XAMPP	Local server (Apache + MySQL)	v8.2+
PHP	Server-side scripting language	v8.1+
MySQL	Relational database management	v8.0+
phpMyAdmin	Database administration GUI	Included in XAMPP
HTML5 / CSS3	Frontend structure and styling	Modern browsers
JavaScript (ES6)	Client-side interactivity	Vanilla JS
VS Code	Primary code editor / IDE	Latest
Google Chrome	Development and testing browser	Latest

3.2 METHODS

The system development follows the Agile methodology, allowing for iterative development and continuous testing across five phases:

1. Requirements Analysis — Gathering functional and non-functional requirements through stakeholder consultation and study of existing hospital workflows.
1. System Design — Designing the database schema (MySQL), system architecture, user interface wireframes, and role-based access control model.

CHAPTER 4 – TECHNOLOGIES

4.1 FRONTEND AND BACKEND PROGRAMMING

4.1.1 *Frontend Programming*

The frontend of GH+ is built using HTML, CSS, and Vanilla JavaScript . The interface follows a red, white, and black design theme reflecting the branding of GH+. CSS custom properties (variables) are used to maintain consistent theming, and all layouts use CSS Grid and Flexbox to ensure full mobile responsiveness across devices of all screen sizes. JavaScript provides client-side interactivity including real-time NIC number validation, tab navigation, modal dialogs, pill-style checkbox and radio button interactions, file upload previews, and an auto-triggering PDF print interface.

4.1.2 *Backend Programming*

The backend is implemented in PHP using a procedural architecture suited to the XAMPP/MySQL environment. All database interactions are performed using MySQLi with prepared statements to prevent SQL injection. Passwords are hashed using PHP's native password hash() with the BCrypt algorithm. Session-based authentication controls access to all protected pages, and every user action is logged to an audit log table in the database. File uploads are validated by file extension and MIME type, and PHP execution is blocked inside the uploads directory via .htaccess for security.

4.2 DATA MODELLING AND MANAGEMENT

The GH+ database (gh_plus) is hosted on MySQL and consists of the following core tables:

- users — Stores all user accounts (patients, doctors, admin) with role, NIC, password hash, and status fields.
- patient details — Stores patient-specific data including diseases and additional medical notes, linked to the users table.
- staff details — Stores medical staff-specific data including position, hospital, department, and SLMC registration number.

- `medical_records` — Stores each medical visit record including diagnosis, prescription, and doctor notes, linking patient and doctor IDs.
- `medical_documents` — Stores metadata for files attached to medical records (images, PDFs, audio, video), with the actual files stored in the `uploads/` directory.
- `support_messages` — Stores patient/staff support enquiries submitted via the Contact Us page.
- `audit_log` — Records all significant user actions including logins, record uploads, approvals, and deletions.

4.2.1 Structure Diagram

The database uses a normalized relational structure. The `users` table serves as the central entity, with `patient_details` and `staff_details` extending it through one-to-one relationships. Medical records link patients and doctors through foreign keys, and documents are linked to records. The audit log references users to track who performed each action. All foreign keys enforce referential integrity with CASCADE DELETE where appropriate.

4.3 IDES AND DEVELOPMENT TOOLS

The primary development environment is Visual Studio Code with extensions for PHP IntelliSense, MySQL, and Live Server. XAMPP Control Panel is used to manage the Apache and MySQL services locally. phpMyAdmin is used for database schema management and query testing. Google Chrome Developer Tools are used for frontend debugging, responsive design testing, and JavaScript console monitoring.

CHAPTER 5 – TIME PLAN

5.1 GANTT CHART

The project follows a structured five-phase development timeline spanning approximately 20 weeks from January to May 2026. Each phase has defined deliverables and milestones to ensure systematic progress.

Task	April	April	May	May	June	June	July	July	
Requirements Analysis									
System & DB Design									
Frontend Development									
Backend Development									
Testing & Bug Fixing									
Documentation & Proposal Writing									
Final Review & Submission									

5.2 SCHEDULE AND BUDGET SUMMARY

5.2.1 Schedule

Project Task	Date	Duration
Requirements Analysis	April 2026	4 Weeks
System & Database Design	April, May - 2026	4 Weeks
Frontend Development	May 2026	4 Weeks
Backend Development	May, June – 2026	6 Weeks
Testing & Bug Fixing	May, June - 2026	6 Weeks
Documentation & Proposal	June, July - 2026	4 Weeks
Final Review & Submission	July - 2026	1 Week

5.2.2 Budget Summary

Item	Duration	Estimated Cost (LKR)
Development Laptop / PC	One-time	0 (Existing)
XAMPP Server Software	Free / Open	0
VS Code IDE	Free	0
Internet Connection (Research)	5 Months	5,000
Printing & Binding (Proposal)	One-time	1,500
Miscellaneous (Stationery etc.)	Project	1,000
Total		7,500

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